

Sealed Experiment Record

Sealed 2026-10-08 · Signed by the K-Veritas server
Session 0b94e254 · Data hash 55b9...cf62

Abstract

This record seals 1 run (kv_run.sh) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-08 01:16 and 01:24 UTC. Reported results: c_nqueen_seconds 2.8733, c_matmul_seconds 1.8112, c_sudoku_seconds 2.104, c_bedcov_seconds 1.1856. 1 invocation in total, 0 failed. Produced with K-Veritas [1].

What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

1 Setup

Item	Value
Session	0b94e254-17e3-4815-8e3f-15890e6cbe6d
Initialized	2026-10-08T01:16:00Z
Machine	167ec0d6dc9a8170
Operating system	linux / amd64
CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
Memory	31.0 GB
GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
Environment, run 1	2d079d310a3777c78784d14a6d98ea3b079170f2dcfb96c5205284e683ebb9e8
Disclosure	open
Source code, run 1	8702e189fbe1dd2a98b68ca455f5a9906ca87f20f6307244ce9e2cc09e15dc75

2 Runs

#	Command	Duration	Exit	Anchored
1	bash kv_run.sh	8 minutes 1 seconds	0	at run end

1 invocation in total: 1 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

3 Results

Run 1: kv_run.sh

Metric	Final value	Step	Source
c_nqueen_seconds	2.8733		explicit
c_matmul_seconds	1.8112		explicit
c_sudoku_seconds	2.104		explicit

Metric	Final value	Step	Source
c_bedcov_seconds	1.1856		explicit
rust_nqueen_seconds	3.0356		explicit
rust_matmul_seconds	1.7426		explicit
rust_sudoku_seconds	2.2547		explicit
rust_bedcov_seconds	1.3404		explicit
go_nqueen_seconds	3.2979		explicit
go_matmul_seconds	2.2567		explicit
go_sudoku_seconds	2.4864		explicit
go_bedcov_seconds	1.5523		explicit
java_nqueen_seconds	4.3004		explicit
java_matmul_seconds	1.8862		explicit
java_sudoku_seconds	3.225		explicit
java_bedcov_seconds	4.5043		explicit
javascript_nqueen_seconds	6.575		explicit
javascript_matmul_seconds	6.626		explicit
javascript_bedcov_seconds	4.9239		explicit
python_nqueen_seconds	132.682		explicit
python_matmul_seconds	146.742		explicit
python_sudoku_seconds	64.173		explicit
python_bedcov_seconds	69.4029		explicit
julia_nqueen_seconds	3.1641		explicit
julia_matmul_seconds	1.8062		explicit
julia_sudoku_seconds	2.8236		explicit
julia_bedcov_seconds	2.2923		explicit

Inline claim	Value	Phase	Output line
c_nqueen_seconds	2.8733		4
c_matmul_seconds	1.8112		7
c_sudoku_seconds	2.104		10
c_bedcov_seconds	1.1856		13

4 Execution evidence

Coherence

Coherent (0.69): the telemetry channels move together, as one process.

Telemetry

Run	Samples	CPU time	Peak memory	Peak GPU
1	4388	147.2 s	0.96 GB	0%

5 Provenance

Run 1: kv_run.sh

Disclosure open. 35 tracked files, 9 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	01:16:01	42e1df7c95306093	35
8	run_end	01:24:02	42e1df7c95306093	0

7 snapshots with no file change omitted here; full timeline in Appendix B.

Activity: 137 files read, 0 files written, 29 subprocesses. Full list in Appendix B.

6 Verification

Field	Value
Signer	Signed by the K-Veritas server
Signed at	2026-10-08T01:24:03.809061+00:00
Algorithm	RSA-PSS-SHA256, 4096-bit key
Key fingerprint	eeac81653cf44ac029c4069cf2a06d9f
Data hash	55b93fb1f5dcea0f3f0e504fc005c1d06d93a4637461dc1f16b38f537f7bcf62
Code bundle hash	29785ffc1967bbff75368401180ea7bc04ad8790a92a3e29cdf6f13e921a9ec5

Run anchors: 1 invocations were anchored at the server when they ended.

Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data_hash:nonce:signed_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

Appendix

A Full metric series

Run 1: kv_run.sh

Metric	Value	Step	Line	Source
c_nqueen_seconds	2.8733		3	explicit
c_matmul_seconds	1.8112		6	explicit
c_sudoku_seconds	2.104		9	explicit
c_bedcov_seconds	1.1856		12	explicit
rust_nqueen_seconds	3.0356		16	explicit
rust_matmul_seconds	1.7426		18	explicit
rust_sudoku_seconds	2.2547		20	explicit
rust_bdcov_seconds	1.3404		22	explicit
go_nqueen_seconds	3.2979		25	explicit
go_matmul_seconds	2.2567		27	explicit
go_sudoku_seconds	2.4864		29	explicit
go_bdcov_seconds	1.5523		31	explicit
java_nqueen_seconds	4.3004		34	explicit
java_matmul_seconds	1.8862		36	explicit
java_sudoku_seconds	3.225		38	explicit
java_bdcov_seconds	4.5043		40	explicit
javascript_nqueen_seconds	6.575		43	explicit
javascript_matmul_seconds	6.626		45	explicit
javascript_bdcov_seconds	4.9239		48	explicit
python_nqueen_seconds	132.682		51	explicit
python_matmul_seconds	146.742		53	explicit
python_sudoku_seconds	64.173		55	explicit
python_bdcov_seconds	69.4029		57	explicit
julia_nqueen_seconds	3.1641		60	explicit
julia_matmul_seconds	1.8062		62	explicit
julia_sudoku_seconds	2.8236		64	explicit
julia_bdcov_seconds	2.2923		66	explicit

Phase	Line	Time	CPU time	Memory	GPU
c	1	01:16:01	0.4 s	0.02 GB	
rust	14	01:16:09	0.1 s	0.02 GB	
go	23	01:16:17	0.3 s	0.02 GB	
java	32	01:16:27	0.4 s	0.05 GB	
javascript	41	01:16:41	0.6 s	0.05 GB	
python	49	01:16:59	0.6 s	0.03 GB	
julia	58	01:23:52	0.6 s	0.14 GB	

B Provenance timeline and activity map

Run 1: kv_run.sh

#	Snapshot	Time	Root	Changed
0	run_start	01:16:01	42e1df7c95306093	35
1	phase c	01:16:01	42e1df7c95306093	0
2	phase rust	01:16:09	42e1df7c95306093	0
3	phase go	01:16:17	42e1df7c95306093	0
4	phase java	01:16:27	42e1df7c95306093	0
5	phase javascript	01:16:41	42e1df7c95306093	0
6	phase python	01:16:59	42e1df7c95306093	0
7	phase julia	01:23:52	42e1df7c95306093	0
8	run_end	01:24:02	42e1df7c95306093	0

Event	Path or command	Hash
read	kv_run.sh	
read	src/c/nqueen	
read	src/c/bedcov.c	
read	src/c/matmul.c	
read	src/c/nqueen.c	
read	src/c/sudoku.c	
read	src/go/bedcov.go	
read	src/go/matmul.go	
read	src/go/nqueen.go	
read	src/go/sudoku.go	
read	src/java/bedcov.java	
read	src/java/matmul.java	
read	src/java/nqueen.java	
read	src/java/sudoku.java	
read	src/javascript/bedcov.js	
read	src/javascript/matmul.js	
read	src/javascript/nqueen.js	
read	src/javascript/sudoku.js	
read	src/julia/bedcov.jl	
read	src/julia/matmul.jl	
read	src/julia/nqueen.jl	
read	src/julia/sudoku.jl	
read	src/python/bedcov.py	
read	src/python/matmul.py	
read	src/python/nqueen-Wirth.py	
read	src/python/nqueen.py	
read	src/python/sudoku.py	
read	src/ruby/bedcov.rb	
read	src/ruby/matmul.rb	
read	src/ruby/nqueen.rb	
read	src/ruby/sudoku.rb	
read	src/rust/bedcov.rs	
read	src/rust/matmul.rs	
read	src/rust/nqueen.rs	
read	src/rust/sudoku.rs	

Event	Path or command	Hash
read	src/scheme/run.sh	
read	src/c/matmul	
read	src/c/sudoku	
read	src/c/bedcov	
read	src/rust/nqueen	
read	src/rust/matmul	
read	src/rust/sudoku	
read	src/rust/bedcov	
read	src/go/nqueen	
read	src/go/matmul	
read	src/go/sudoku	
read	src/go/bedcov	
read	src/java/nqueen.class	
read	src/java/matmul.class	
read	src/java/sudoku.class	
read	src/java/bedcov.class	
read	src/java/Interval.class	
read	src/java/StackCell.class	
read	.gitignore	
read	LICENSE	
read	README.md	
read	analysis/Makefile	
read	analysis/parse-readme.pl	
read	analysis/plot.gp	
read	src/c/Makefile	
read	src/cl/Makefile	
read	src/cl/matmul.asd	
read	src/cl/matmul.lisp	
read	src/cl/nqueen.asd	
read	src/cl/nqueen.lisp	
read	src/cl/sudoku.asd	
read	src/cl/sudoku.lisp	
read	src/crystal/Makefile	
read	src/crystal/bedcov.cr	
read	src/crystal/matmul.cr	
read	src/crystal/nqueen.cr	
read	src/crystal/sudoku.cr	
read	src/csharp/Makefile	
read	src/csharp/bedcov.cs	
read	src/csharp/bedcov.csproj	
read	src/csharp/matmul.cs	
read	src/csharp/matmul.csproj	
read	src/csharp/nqueen.cs	
read	src/csharp/nqueen.csproj	
read	src/csharp/sudoku.cs	
read	src/csharp/sudoku.csproj	
read	src/d/Makefile	
read	src/d/bedcov.d	
read	src/d/matmul.d	
read	src/d/nqueen.d	

Event	Path or command	Hash
read	src/d/sudoku.d	
read	src/dart/Makefile	
read	src/dart/bedcov.dart	
read	src/dart/matmul.dart	
read	src/dart/nqueen.dart	
read	src/dart/sudoku.dart	
read	src/elixir/matmul.exs	
read	src/elixir/nqueens.exs	
read	src/elixir/nqueens.maps.exs	
read	src/fortran/Makefile	
read	src/fortran/matmul.F90	
read	src/fortran/nqueen.F90	
read	src/fortran/sudoku.F90	
read	src/go/Makefile	
read	src/java/Makefile	
read	src/lua/bedcov.lua	
read	src/lua/matmul.lua	
read	src/lua/nqueen.lua	
read	src/lua/sudoku.lua	
read	src/mojo/Makefile	
read	src/mojo/matmul.mojo	
read	src/mojo/nqueen.mojo	
read	src/nim/Makefile	
read	src/nim/bedcov.nim	
read	src/nim/matmul.nim	
read	src/nim/nqueen.nim	
read	src/nim/sudoku.nim	
read	src/ocaml/Makefile	
read	src/ocaml/matmul.ml	
read	src/ocaml/nqueen.ml	
read	src/perl/matmul.pl	
read	src/perl/nqueen.pl	
read	src/perl/sudoku.pl	
read	src/php/matmul.php	
read	src/php/nqueen.php	
read	src/python/Makefile	
read	src/rust/Makefile	
read	src/scheme/matmul.scm	
read	src/scheme/nqueens.scm	
read	src/swift/Makefile	
read	src/swift/bedcov.swift	
read	src/swift/matmul.swift	
read	src/swift/nqueen.swift	
read	src/swift/sudoku.swift	
read	src/v/Makefile	
read	src/v/bedcov.v	
read	src/v/matmul.v	
read	src/v/nqueen.v	
read	src/v/sudoku.v	
read	src/zig/Makefile	

Event	Path or command	Hash
read	src/zig/matmul.zig	
read	src/zig/nqueen.zig	
process	python3 -	
process	./nqueen	
process	./matmul	
process	./sudoku	
process	./bedcov	
process	./nqueen	
process	./matmul	
process	./sudoku	
process	./bedcov	
process	./nqueen	
process	./matmul	
process	./sudoku	
process	./bedcov	
process	java nqueen	
process	java matmul	
process	java sudoku	
process	java bedcov	
process	node nqueen.js	
process	node matmul.js	
process	node sudoku.js	
process	node bedcov.js	
process	python3 nqueen.py	
process	python3 matmul.py	
process	python3 sudoku.py	
process	python3 bedcov.py	
process	julia nqueen.jl	
process	julia matmul.jl	
process	julia sudoku.jl	
process	julia bedcov.jl	

C Run history

Invocation 1, OK, 8 minutes 1 seconds, 66 output lines, started 2026-10-08T01:16:00

metric_hash 580d2046476424c4d54454222a6f65223d4a42c29a96420ef4bea87ed45a92cb
run_digest 6ccceaf0bfaef94500382564fb387696dffa86639f1d642d2eac8ede843b47c6
anchored at run end

D Raw seal fields and hashes

Seal

data_hash 55b93fb1f5dcea0f3f0e504fc005c1d06d93a4637461dc1f16b38f537f7bcf62
signed_message_hash a9f59e1c322489a4d0393ead2b039f030179e948171d17978e14363ec06de73f
nonce f2d058f3707cb17102cba3db10af8c85
signed_at 2026-10-08T01:24:03.809061+00:00
sealed_at 2026-10-08T01:24:03.907264344Z
checkout_bundle 29785ffc1967bbff75368401180ea7bc04ad8790a92a3e29cdf6f13e921a9ec5

Signature (base64)

PgC7Am8M3b4k50hJseIzvBNsa3ShZmmzde11t/nov94ciEx5V2PUCfYV34T5WGSi0EksGM74Sncc0Ue7qTxreat
 KfG15/oNN3/jJ3ICrFv04RjLHHU543z01PjeUpxCCl+vzuA9jy1Aq3NNPhasXjecaq+LGcFCg0fdUHHA1HiG0fwy
 2yGqQMxLZV50GdcUYObRUuioGo76Q+stcAoJDZa+ak9IvHlh58sN2eLkv4287wMrFFQ6fHj08B6zCV1N5nv4pIZ6
 i/1G8CXDBSR6KGLsRPFy84K18HU6fwfI0gh9BwJTFw0KKMDki85l7tZJqaesaNnQ17TMuX5XEr8ajJwK6ioysx6d
 ZhF0lStTcfrNVZ4BpmKl0JUDL1v3AxjN3RNrwuyh0T7ZM5kZrdZGQoI8QQY1uKeK3LlvodN6vYa7CXFu3zTLEKSc
 buYIxgjjANT/r6ggYkrejdXCEChS5x0Pl20eSZnn1DxCkSiwz0BbinX7cvChjC9w7McGuv9QW98zAH0WD65V64
 8e2v2Ue2KgXFnXiFzjFBq59r9G0ULAW/LEnozJ0gfmKQbeDr4S62PYA8dxGJk+Vj6FdVnkfCjVXXSltEsvB2Znc6
 c0zaERtHzpgt7zVtYXNI9saUtAtguHrZVRP9Fi62IAEsriOE5enJzU0hIE2FGZJba48=

Public key

-----BEGIN PUBLIC KEY-----

MIICIjANBgkqhkiG9w0BAQEFAAOCAg8AMIICCgKCAgEA1QL3YyMgeShx8IhTC7jb
 sM0w2BnRXgsBn3+SMXKjtwHATqdekmlPlcjayJxD5fcPrOPPCqp/LouFz5CUZzRi
 6+8FTv70dHuIz5A4Lat4vcgwP+iWscPc3HzpmjpsKbTm0emqgp6RB6GNNPpxYa5X
 7qU95EFEEisX/GfoS9ToXtqPBwCcVwSPBX8efec4TC58m96Fsp4mC8KIcEadbIbHN
 av0JjdTJ//CMOWZSIbdzE6qvEbNFCqQB5tNU7oHfYvBeNLrvPU9TAqdbtozdhQ+D
 AhfTRuJNgWEkjwD5lb0Vhi4qlkweSOEMTKT3ye9IL5xEaJn4bku4eKqeS7VSZt3o
 XbUcmr2H5mKcKsvt6bXmX6ajAmgi900Cm9QJlaQkJM6z2iITnyvDMgueJtFxQpC9
 HNBPgPZ242NH8JgE6CNF7ToL9syasS67u7UMgIcNgeNHNcy9bLgpU5annzP9gN2l
 6HEMLdkXNDKQRb1rDwyEJf3N/50YHnXXPmuf5sb8R/pxwNqtF04R5UGVV92qd2Mi
 1GN7LE3mbQRbPsLL/JapDXMNzNikaFfBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg
 +eE4bf/olCHBCf1VIDCkDxoehdZ0WRRcgM9iK3I/0zZdaBsgdc8WuJRMb8LPP9v
 8Jap/1PwjNcNku5cdW3gHLCCAWEAAQ==

-----END PUBLIC KEY-----

Run 1 outputs

```
stdout d98cbad7679a61c25462141c975148ef713d2b05bd0752321760f61955b40c29
stderr e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics 580d2046476424c4d54454222a6f65223d4a42c29a96420ef4bea87ed45a92cb
digest 6ccceaf0bfaef94500382564fb387696dffa86639f1d642d2eac8ede843b47c6
kv_run.sh (unchanged)
pre 391462af7ac27c7fd0e6001521e76188f49c540897a30937922b7904c87cbbfe
post 391462af7ac27c7fd0e6001521e76188f49c540897a30937922b7904c87cbbfe
```

E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},
  author={Mamadou K. Keita and Christopher Homan},
  year={2026},
  eprint={2605.08586},
  archivePrefix={arXiv},
  primaryClass={cs.CR},
  url={https://arxiv.org/abs/2605.08586},
}
```